Color Indices for Weed Identification Under Various Soil, Residue, and Lighting Conditions

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Abstract. *Color slide images of weeds among various soils and residues were digitized and analyzed for red, green, and blue (RGB) color content. Red, green, and blue chromatic coordinates (rgb) of plants were very different from those of background soils and residue. To distinguish living plant material from a nonplant background, several indices of chromatic coordinates were studied, tested, and were successful in identifying weeds. The indices included r-g, g-b, (g-b)/|r-g|, and 2g-r-b. A modified hue was also used to distinguish weeds from non-plant surfaces. The modified hue, 2g-r-b index, and the green chromatic coordinate distinguished weeds from a nonplant background (0.05 level of significance) better than other indices. However, the modified hue was the most computationally intense. These indices worked well for both nonshaded and shaded sunlit conditions. These indices could be used for sensor design for detecting weeds for spot spraying control.* **Keywords.** *Weeds, Optics, Color, Sprayers.*

Optical detection of plants has been proposed as a means to control the operation of an intermittent, or spot sprayer (Haggar et al., 1983; Von Bargen et al., 1993; Felton and McCloy, 1992; Nelson, 1993; and Mortensen et al., 1992). Microprocessor technology permits the use of optical sensors to detect and spray individual target plant pests (weeds). Intermittent spraying, or spot spraying, is the application of pesticides only in the presence of the pest. Selected combinations of wavelength bands can be used to distinguish plants from soil background (Haggar et al., 1984; Franz et al., 1991a). Electronic silicon-based sensors with appropriate optical filters for red and near-infrared (NIR) light have been used to identify the presence of plants against soil backgrounds (Haggar et al., 1984; Von Bargen et al., 1993). In designing a simple optical plant sensor for a spot sprayer, a reflectance and field-of-view analysis is performed to distinguish the plant from its natural background which includes the soil and plant residue, each with unique reflectance characteristics (Woebbecke, et al., 1994). Optical filters are selected based on plant and background reflectance characteristics (Haggar et al., 1983; Nitsch et al., 1991; Von Bargen et al., 1993). Using just the information of the presence or absence of a weed as input to control a spot sprayer, a significant reduction in herbicide use is possible especially where weeds exist in nonuniform field situations, with considerable cost savings (Johnson et al., 1995; Felton and McCloy, 1992). Definitions of optical terms are given in The Photonics Dictionary™ (1993).

Guyer et al. (1986) distinguished plants from a soil background by illuminating plants with radiation containing a high ratio of near-infrared to visible wavelengths. When these plants were viewed using a charge injection device (CID) camera, the plant picture had elements (pixels) with greater intensities than soil pixels. However, near-infrared reflectance was similar for plant and highly reflective wheat straw residue (Nitsch, 1991). Residue has been successfully discriminated from soil backgrounds using spectral analysis and cameras sensitive to near-infrared and red light (Meyer et al., 1988).

Color has been used to distinguish plants from their natural background. Color is usually based on the red, green, blue (RGB) primary system. This RGB color model is derivable from spectral measurements (Wyszecki and Stiles, 1982). Once the RGB color primaries are identified, other colors in the model can be identified or matched (Jain, 1989). Thomas et al. (1988) used 35 mm color slides of canopy-soil combinations and transformed these color images into gray-scale renditions of each primary color. This was accomplished by viewing each slide through separate red, green, and blue colored filters mounted on a video camera. By subtracting the red image from the green or blue image, a gray-scale image resulted with perceptually brighter leaf pixels and darker soil pixels. Meyer and Davison (1987) used both color and black and white infrared film, with Wratten filters to distinguish the soybean canopy from soil and residue. The best distinction occurred using color infrared film. Color images of residue were used to determine the percent of soil covered with residue (Morrison and Chichester, 1991).

Tarbell and Reid (1991) characterized the color of corn leaves using RGB coordinates. By creating slide images with different techniques, they found that values of RGB coordinates could be quite variable based on benchmark color squares located within each image. Therefore, chromaticity coordinates were used, providing more useful results than RGB coordinates. Color co-occurrence

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Mention of specific trade names is for information only and not to the exclusion of others that might be suitable.

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matrices have been used to identify the textural features of plants (Shearer and Holmes, 1990). Textural features analysis requires that the target pixels be compared with neighboring pixels, a time-consuming operation. Hue, saturation, and intensity (HSI) can also describe the color of an object (Gonzalez, 1992).

Guyer et al. (1986) used image shape feature analysis to distinguish between plant species. Two problems must be solved in detection, identification, and enumeration of individual plants using image analysis. The first is that plants must be distinguishable from the soil and residue background in the image. As light conditions change, the perceived color of an object will also change. A binary or high contrast image must be developed from the original color image. Secondly, plant species identification has been traditionally performed using leaf shape analysis on binary images. Individual leaves in these images may be partially occluded by other parts of the canopy, through physical hiding or stacking of leaves or by shadows cast by one leaf onto another (Franz et al., 1991b). It would be advantageous to discern between monocots and dicots based on whole plant shape. Most post-emergent herbicides are selective in controlling one plant type or the other. Individual species identification might not be necessary at this time.

This article addresses the problem of segmenting plants from soil and residue by studying optical and colorimetric considerations for distinguishing plants using 35-mm color slides as source information. The specific objectives of this study were to:

- Determine a color index method from the RGB primary colors or HSI which best distinguishes plant material from their background, from various color images of weed, soil, residue, and background light conditions including shaded and unshaded plant and soil surfaces.
- Determine whether color indices represent a possible method for classifying plant species as either a monocot or dicotuse
- Use these indices to obtain a binary image, segmenting plants from their background for future plant identification analyses.

Materials and Methods

The non-normalized RGB coordinates of individual weed species in combination with various soils and residues were acquired from 312 photographic slides. Weed species included common cocklebur (*Xanthium strumarium* L.), velvetleaf (*Abutilon theophrasti* Medicus), field bindweed (*Convolvulus arvensis* L.), and large crabgrass (*Digitaria sanquinalis* (L.) Scop.). The RGB color primaries were obtained by digitally scanning these slides.

Color Slide Information

Color slides (35 mm) of these plant species were acquired under natural lighting in a glass greenhouse in Lincoln, Nebraska. A single layer of polypropylene Saran® cloth screening provided 55% shading. Outdoor solar radiation ranged from 530 to 635 W/m^2. Photographs were taken directly above the plants using a Nikkormat camera, with a 55-mm lens (fig. 1). The camera was mounted

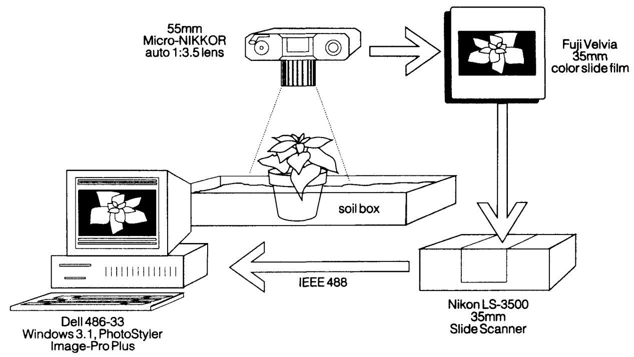

Figure 1–Acquisition and processing of plant images for color separation and species identification using the 35-mm slide film method.

600 mm above the background surface. A uniformly focused image resulted with a depth of field from 500 to 760 mm. Color slide film with a speed of 50 ASA was used. The camera f-stop was adjusted between 5.6 and 11, while the shutter speed was varied from 1/30 to 1/60 s.

Plants were one month old, except for the large crabgrass which was 2 1/2 weeks. Projected canopy profile area of common cocklebur and velvetleaf captured within the slides ranged from 3000 to 5000 mm^2 and 1500 to 2500 mm^2, respectively (in comparison, a quarter coin would be approximately 800 mm^2). The height for both plant species was approximately 100 to 150 mm. One of the two field bindweed plants which eventually lodged (collapsed under its own weight onto the soil surface) had as length of 230 mm and projected area of 1200 mm^2. The other bindweed remained erect and had a projected area of 600 mm^2 and a height of 70 mm. Large crabgrass plants had a projected area of 1000 to 1500 mm^2, and a height of around 130 mm.

Several field conditions were simulated by placing residue and soil under the plants. Wheat and corn residue and Sharpsburg soil (fine, montmorillonitic, mesic; subgroup; typic argiudol; of the order mollisol) were used. The hue, saturation, and intensity of the soil was found to be 10YR4/2 (dark grayish brown) dry and 10YR2/2 (very dark brown) wet, using a traditional Munsell soil color chart. Residue was positioned about the plant so that approximately 80 to 90% of the soil surface was covered. The color of the residue ranged from yellow-brown to light grayish-brown, due to the orientation of the residue with respect to the camera, incident light, and degree of weathering. Residue was categorized according to type (corn or wheat residue) and color (yellow or white). As the plant developed, upper leaves tended to shade lower leaves along with the soil and residue beneath the plant. The color of shaded surfaces was naturally darker than when unshaded.

Other sets of images were taken of potted large crabgrass (one month old) and field bindweed (1 1/2 months old) with outdoor solar radiation levels at 550 W/m^2 and the screening removed. These crabgrass plants were 200 to 300 mm in height with projected areas of 3000 to 6000 mm^2. Field bindweed had a 1000 to 2000 mm^2 projected area with vine lengths ranging from 100 to 240 mm. The soil background consisted of a dry Steinauer loamy sand (fine-loamy, mixed (calcareous), mesic, typic udorthent) and had a Munsell color of 2.5Y8/2 (white).

Digital RGB Color Images

After film development, each slide was scanned digitally using a Nikon, LS-3500 scanner (fig. 1). The scanner was interfaced to a Dell 486-33MHz computer with 16 megabytes of memory using a National Instruments GPIB board. The scanning software was Aldus Photostyler® under Microsoft Windows 3.1®. The resultant scanned image was 24-bit color (8 bits red, 8 bits green, and 8 bits blue), each representing a non-normalized RGB coordinate with a decimal value from 0 to 255. The image data using the TIFF format was saved on disk. This format is available with many commercial systems. Because each image varied in their quality and brightness, the scanning process was adjusted so that the digital image most closely resembled the original plant scene on the slide. Scanner exposure levels were set between 20 and 50 and a black level of zero. The color balance of each red, green, and blue primary were all set at 50, as to provide medium intensity, but not favor one primary over another. The gamma adjustment which controls midtone colors was set to the nonlinear mode with a default value of one.

Digital Image Analysis

The image processing program, Image Pro Plus v2.0® Inc., was used to study and evaluate individual red, green, and blue components (8 bits each) of the color image. Target surfaces (plant, soil, or residue) were analyzed and identified using the area of interest (AOI) tool with the mouse pointing device. This tool was used to determine the statistical or optical density of the AOI. The average value for each non-normalized coordinate (R_c, G_c, and B_c) was computed, based on at least 100 pixels within the AOI. These average values were then collectively averaged with other average coordinates taken at other image locations containing the same surface type. Non-normalized RGB coordinates are directly proportional to total light reflected from a surface. They are highly sensitive to the intensity of the illuminating source and its angle with the target surface. Non-normalized RGB coordinates cannot be used to make conclusions about the spectral reflectance of an object, only the color (Gonzalez and Woods, 1992).

Chromatic Coordinates

The red, green, and blue chromatic coordinates or chromaticity coordinates (r, g, and b) were used to derive colors for the HSI model, (Tarbell and Reid, 1991). Chromatic coordinate values are defined as:

$$r = \frac{R}{R+G+B},\ g = \frac{G}{R+G+B},\ b = \frac{B}{R+G+B} \quad (1)$$

where R, G, and B are the normalized RGB coordinates ranging from 0 to 1 (fig. 2). Normalized RGB coordinates were obtained simply as:

$$R = \frac{R_c}{R_m},\ G = \frac{G_c}{G_m},\ B = \frac{B_c}{B_m} \quad (2)$$

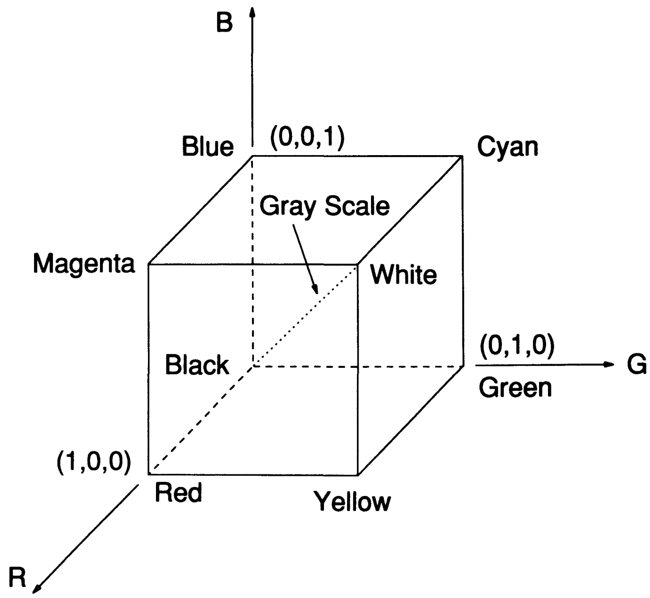

Figure 2–The RGB color model. R, G, and B are normalized and within the range [0, 1].

For the color display system chosen, R_m, G_m, and B_m are maximum non-normalized RGB values. Because different color display systems will have different ranges of values, equation 2 must always be used with equation 1. For example, an indexed 256 color display may use a 3-3-2 bit RGB system or an index pointing to a true color based on a RGB system. Since $R_m = G_m = B_m = 255$ for 24-bit color images, these two equations combined give:

$$r = \frac{R_c}{R_c + G_c + B_c},\ g = \frac{G_c}{R_c + G_c + B_c},$$

$$b = \frac{B_c}{R_c + G_c + B_c} \quad (3)$$

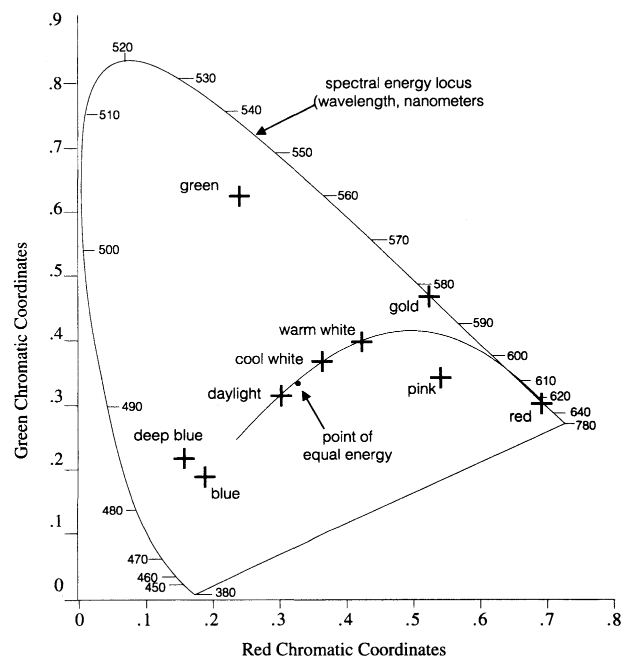

Figure 3–CIE chromaticity diagram.

Color coordinates are frequently plotted in a chromaticity diagram (fig. 3). This diagram characterizes the composition of a color as a function of red and green. The corresponding blue chromatic coordinate can be found from:

$$r + g + b = 1 \qquad (4)$$

MODIFIED HUE

The perceptual attributes of color are commonly referred to as hue, saturation, and intensity (HSI) as shown in figure 4. The HSI model is a cylindrical coordinate system with red, green, and blue spaced 120° apart, and intensity along the vertical axis. Saturation is the distance measured outward along the vertical axis and is the degree to which a pure color is diluted by white light. Hue is the angle measured from the red axis to the point of interest and describes pure color. Hue is a good descriptor of the "redness", "greenness", and "blueness" of an object and is calculated from the non-normalized RGB coordinates as:

Hue =

$$\cos^{-1}\left[\frac{2R_c - G_c - B_c}{2\left[R_c^2 + G_c^2 + B_c^2 - R_cG_c - G_cB_c - R_cB_c\right]^{1/2}}\right] \qquad (5)$$

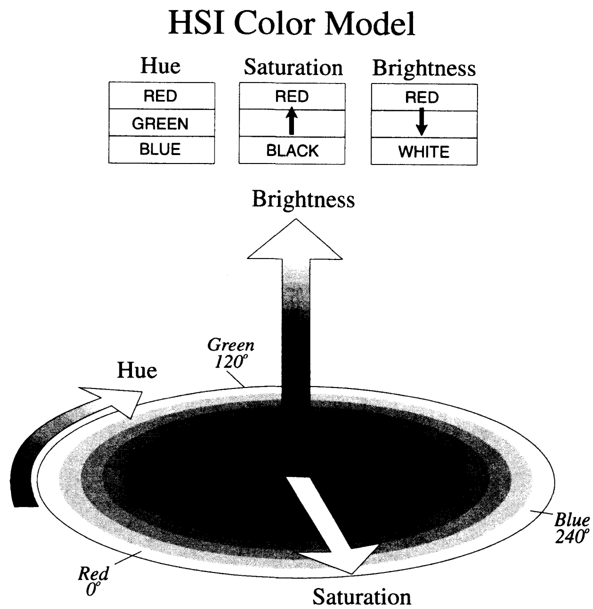

Figure 4–The HSI color model.

However, equation 5 implies that if $B_c > G_c$, then hue = 360° – hue, since the inverse cosine specifies an angle between 0° and 180°. The angular notation of hue presents difficulties. Problems arise when a point lies either to the left or right of the starting point on the red axis. Although

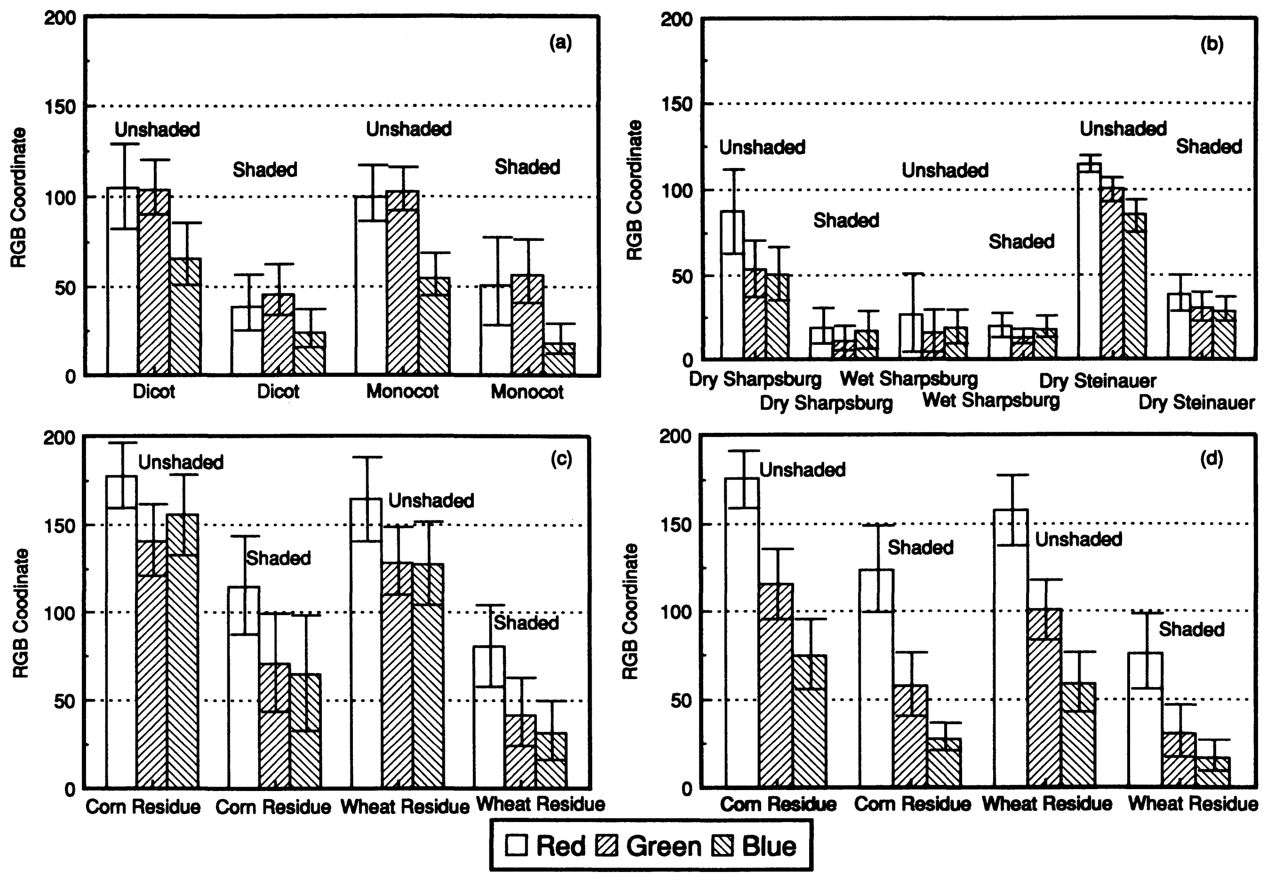

Figure 5–Comparison of RGB coordinates for determining optical contrast of (a) plants, (b) soils, (c) white residue, and (d) yellow residue. Vertical lines represent 1 standard deviation from the mean.

the hue of any two objects may only vary 2°, the hue of the first object might be 1° while the other was 359°. In fact, they have nearly the same hue. To eliminate this possibility, hues greater than 180° were represented as negative angles. Thus, the modified hue had a working range from – 180° to + 180°. During the study, no value of hue was found within the range of 146° to 228° for any field surface. The modified hue of green was therefore + 120° (the same as the hue) and – 120° for blue (compared to + 240° for hue).

Optical Contrast Indices

To find the best optical contrast between plants and their background, the following indices were selected and evaluated:

Subtracting the green chromatic coordinate from the red chromatic coordinate (r-g) (6a)

Subtracting the blue chromatic coordinate from the green chromatic coordinate (g-b) (6b)

Dividing the first two indices $(g-b)/|r-g|$ (6c)

Subtracting the first two indices in the form 2g-r-b (6d)

To eliminate the possibility of obtaining a zero in the denominator of the index $(g-b)/|r-g|$, denominator values between – 0.01 and + 0.01 were set to 0.01. Some leaves have a much larger green than the red chromatic coordinate. This resulted in the index $(g-b)/|r-g|$ having a small value, which was undesirable. When (r-g) was less than – 0.08, (r-g) was fixed at 0.01. Modified hue was also evaluated.

Statistical Tests

The optical contrast between the plant and the background was statistically analyzed using the student's t-test on the difference between the means of each surface, at the 0.05 level of significance (Woebbecke, 1993). It was assumed that the non-normalized RGB coordinates were normally distributed over the surfaces. The null hypothesis indicated there was no color or index differences among different surfaces. A pairwise comparison between shaded and unshaded surfaces was made to determine if color changed significantly due to shading. Finally, a test of color between monocot and dicot weeds was performed. If significant, it would not be necessary to continue future work with a shape feature analysis.

Contrast Indices Applied to Whole Images

Image-Pro provided the ability to analyze 24-bit images for some of the above analyses with the existing computer equipment. However, 8-bit indexed 256 color images were also tested and worked satisfactorily. The 24-bit images were converted to the 8-bit images using the software Leadview® v2.6. This commercial software automatically computed 256 of the most frequent colors and dithered the rest using a nearest-color algorithm. Thus, this color subset remained true to the original image. A program was written with the HALO® Professional Graphics Kernel System to calculate the chromatic coordinates, contrast indices, and the modified hue on each plant-soil-residue image, pixel by pixel (Woebbecke, 1993). After the contrast indices were calculated on an entire 8-bit image, the resulting near-binary image was visually examined to identify areas where the best and poorest contrast had occurred. Pixels with a low intensity ($R_c + G_c + B_c < 40$) were automatically zeroed, because any variation in a small, non-normalized RGB coordinate value could result in a large variation in the chromatic coordinates. A histogram equalization was performed on each near-binary image to enhance the final visual contrast (Gonzalez and Woods, 1992).

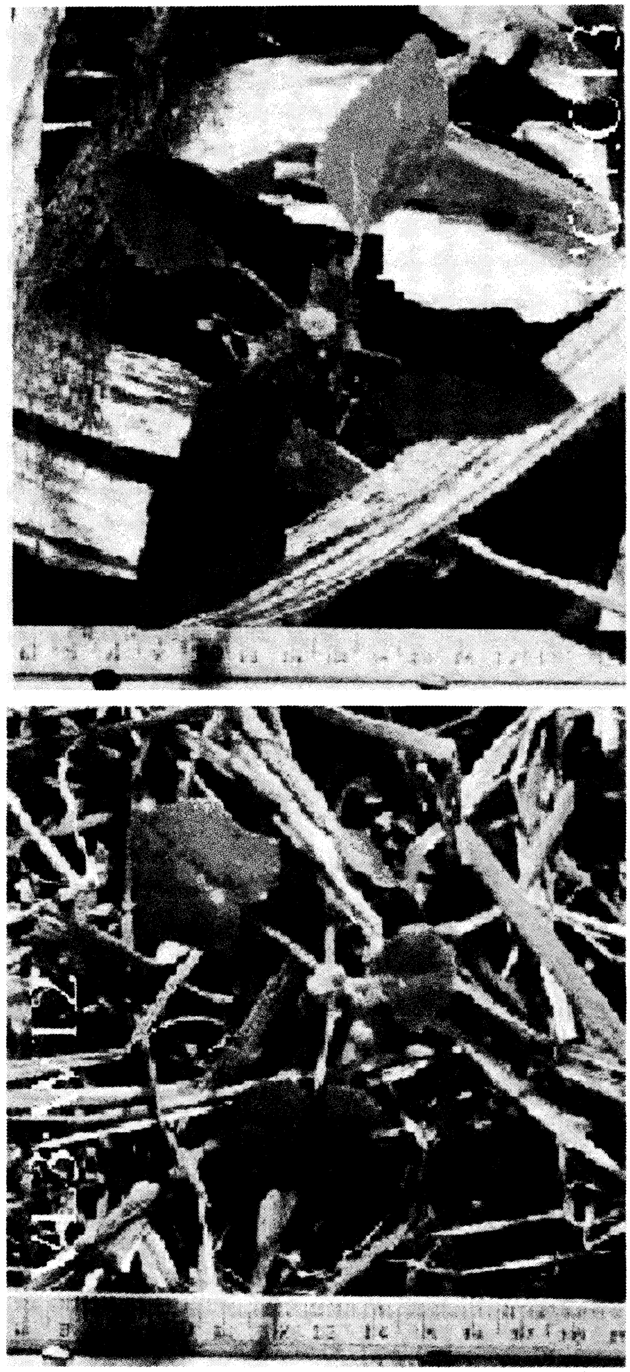

Figure 6–Color photographic images of common cocklebur and velvetleaf in various residues.

Results and Discussion

Use of Non-normalized RGB Coordinates

The RGB coordinates can be derived theoretically from any spectral radiance curve provided that the tristimulus vectors $\bar{x}$ [λ(wavelength)], $\bar{y}$ (λ), and $\bar{z}$ (λ) are available from the manufacturer of the film or video camera (Wyszecki and Stiles, 1982). If spectral reflectance could be used for optical classification, many of the previous problems associated with classifying surfaces based on non-normalized RGB coordinates would be reduced or eliminated. Unfortunately, manufacturers are reluctant to release this information and spectral information cannot be derived directly from the tristimulus vectors. These vectors were indirectly measured with the Photostyler® program.

The non-normalized RGB coordinates (R_c, G_c, and B_c in eq. 2) for a given plant or soil surface were very similar (S.D. = 7 on a 0 to 255 scale). When these coordinates were compared to other non-normalized RGB coordinates from other images, considerable variation was found (S.D. = 33 on a 0 to 255 scale), similar to results found by Tarbell and Reid (1991). This was due to variability in lighting at the time the slide was taken and possibly the film development process. Shaded surfaces had lower values than their unshaded counterparts (fig. 5).

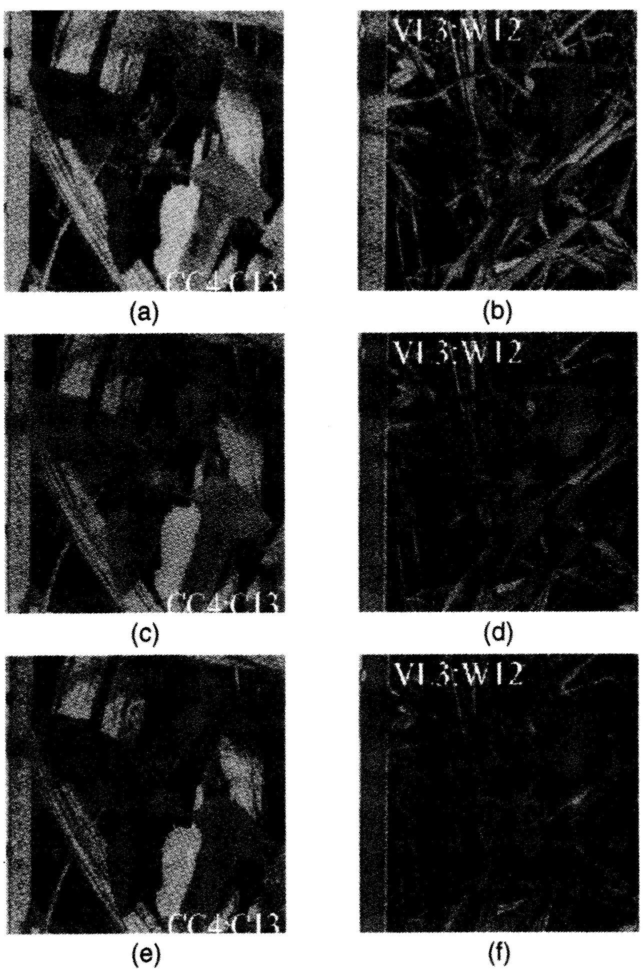

Figure 7–Red (a, b), green (c, d) and blue (e, f) coordinate values of a cocklebur in corn residue (left) and a velvetleaf in wheat residue (right).

Table 1. Mean chromatic coordinates and contrast indices and their statistical significance for unshaded and shaded field surfaces

Surface (Unshaded/ shaded)	Chromatic Coordinate r	g	b	Contrast Indices r-g	g-b	$\frac{g-b}{\lvert r-g\rvert}$	2g-r-b	Mod Hue
Dicot	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/
	0.35*	0.43*	0.22*	–0.08*	0.21*	14.95*	0.29*	85*
Monocot	0.39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/
	0.40	0.46*	0.14*	–0.06*	0.31*	20.62*	0.38*	72*
Dry Sharpsburg	0.46/	0.28/	0.26/	0.18/	0.01/	0.07/	–0.17/	8/
Soil	0.43	0.23*	0.34*	0.20	–0.10*	–2.64*	–0.30*	42*
Wet Sharpsburg	0.40/	0.24/	0.36/	0.15/	–0.12/	–2.59/	–0.27/	48/
Soil	0.37	0.25	0.38	0.12	–0.13	–6.18	–0.25	51
Dry Steinauer	0.38/	0.33/	0.28/	0.05/	0.05/	1.10/	0.00/	30/
Soil	0.39	0.31*	0.30	0.08*	0.01*	0.06*	-0.06*	25*
White Corn	0.38/	0.30/	0.33/	0.08/	–0.03/	–0.42/	–0.11/	25/
Residue	0.47*	0.28*	0.25*	0.19*	0.03*	0.12*	–0.16*	8*
White Wheat	0.39/	0.31/	0.30/	0.08/	0.00/	0.04/	–0.08/	8/
Residue	0.53*	0.27*	0.20*	0.27*	0.07*	0.26*	–0.20*	11*
Yellow Corn	0.48/	0.31/	0.20/	0.17/	0.11/	0.69/	–0.06/	24/
Residue	0.60*	0.27*	0.13*	0.33*	0.14*	0.44*	–0.19*	17*
Yellow Wheat	0.50/	0.32/	0.18/	0.19/	0.13/	0.78/	–0.05/	25/
Residue	0.63*	0.24*	0.13*	0.39*	0.11*	0.32*	–0.28*	13*

* Mean values are significantly different between the shaded and unshaded surface at the 0.05 level of significance.

The red and green coordinates extracted from the color image shown in figure 6 were nearly the same for most plant leaves (fig. 7a, b, c, d). Residue had a slightly lower (darker) green coordinate than the red. In the blue coordinate, figure 7e, f was lower for the plant than the other two colors, while the blue coordinate of the unshaded

Table 2. Monocot and dicot mean chromatic coordinates and contrast indices and their statistical significance

Surface	Chromatic Coordinate r	g	b	Contrast Indices r-g	g-b	$\frac{g-b}{\lvert r-g\rvert}$	2g-r-b	Mod Hue
Unshaded Dicot/	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/
Unshaded Monocot	0.39	0.40*	0.21*	–0.01	0.19*	10.46*	0.20*	63
Unshaded Dicot/	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/
Shaded Monocot	0.40	0.46*	0.14*	–0.06*	0.31*	20.62*	0.38*	72*
Shaded Dicot/	0.35/	0.43/	0.22/	–0.08/	0.21/	14.95/	0.29/	85
Unshaded Monocot	0.39*	0.40*	0.21	–0.01*	0.19	10.46*	0.20*	63*
Shaded Dicot/	0.35/	0.43/	0.22/	–0.08/	0.21/	14.95/	0.29/	85/
Shaded Monocot	0.40*	0.46*	0.14*	–0.06	0.31*	20.62*	0.38*	72*

* Mean values are significantly different from each other at the 0.05 level of significance.

white residue was as bright as the green coordinate values. The shaded residue was darker. Non-normalized RGB coordinates could not distinguish between plant and non-plant surfaces nor classify plants as monocots or dicots.

Chromatic Coordinates

The normalized chromatic coordinates for shaded and unshaded surfaces were not as variable as non-normalized RGB coordinates. For example, the percent change in the green chromatic coordinate was 13% for shaded and unshaded white wheat residue, while the percent change in the non-normalized green coordinate was 67%. The red, green, and blue chromatic coordinates for shaded and unshaded surfaces were statistically different (table 1). Generally, chromatic coordinates provided a better representation of the color of a surface.

Chromatic coordinates of monocot and dicot plants were similar (fig. 8a) although the green chromatic coordinate was statistically significant in classifying monocots and dicots (table 2). However, color differences were probably the result of changes in incident light, camera angle, or plant age. The green chromatic coordinates for plant leaves differed significantly from those of soil and residue surfaces, for both unshaded and shaded (tables 3 and 4) conditions. The red and green plant chromatic coordinate values were nearly the same (0.35 to 0.45) while the blue chromatic coordinate had a lower proportion of the total intensity (0.15 to 0.25) (fig. 8a).

Soil and residue had a higher proportion of red than either of the other two primary colors (fig. 8). In general, nonplant surfaces had red chromaticities greater than or equal to those of plant surfaces. The green chromaticities

Table 3. Unshaded field surfaces mean chromatic coordinates and contrast indices and their statistical significance

	Chromatic Coordinate			Contrast Indices				Mod Hue
Surface Pair	r	g	b	r-g	g-b	(g-b)/|r-g|	2g-r-b	
Dicot/Dry	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/8*
Sharpsburg Soil	0.46*	0.28*	0.26*	0.18*	0.01*	0.07*	–0.17*	
Dicot/Wet	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/48*
Sharpsburg Soil	0.40	0.24*	0.36*	0.15*	–0.12*	–2.59*	–0.27*	
Dicot/Dry	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/30*
Steinauer Soil	0.38	0.33*	0.28*	0.05*	0.05*	1.10*	0.00*	
Dicot/White	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/25*
Corn Residue	0.38	0.30*	0.33*	0.08*	–0.03*	–0.42*	–0.11*	
Dicot/White	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/8*
Wheat Residue	0.39*	0.31*	0.30*	0.08*	0.00*	0.04*	–0.08*	
Dicot/Yellow	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/24*
Corn Residue	0.48*	0.31*	0.20*	0.17*	0.11*	0.69*	–0.06*	
Dicot/Yellow	0.38/	0.38/	0.24/	0.00/	0.14/	7.84/	0.14/	62/25*
Wheat Residue	0.50*	0.32*	0.18*	0.19*	0.13*	0.78*	–0.05*	
Monocot/Dry	0.39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/8*
Sharpsburg Soil	0.46*	0.28*	0.26*	0.18*	0.01*	0.07*	–0.17*	
Monocot/Wet	0.39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/48*
Sharpsburg Soil	0.40	0.24*	0.36*	0.15*	–0.12*	–2.59*	–0.27*	
Monocot/Dry	0.39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/30*
Steinauer Soil	0.38	0.33*	0.28*	0.05*	0.05*	1.10*	0.00*	
Monocot/White	0–39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/25*
Corn Residue	0.38*	0.30*	0.33*	0.08*	–0.03*	–0.42*	–0.11*	
Monocot/White	0.39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/8*
Wheat Residue	0.39	0.31	0.30	0.08	0.00	0.04	–0.08	
Monocot/Yellow	0.39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/24
Corn Residue	0.48*	0.31*	0.20*	0.17*	0.11*	0.69*	–0.06*	
Monocot/Yellow	0–39/	0.40/	0.21/	–0.01/	0.19/	10.46/	0.20/	63/25*
Wheat Residue	0.50*	0.32*	0.18*	0.19*	0.13*	0.78*	–0.05*	

* Mean values are significantly different from each other at the 0.05 level of significance.

Table 4. Shaded field surfaces mean chromatic coordinates and contrast indices and their statistical significance

	Chromatic Coordinate			Contrast Indices Indices				Mod Hue
Surface Pair	r	g	b	r-g	g-b	(g-b)/|r-g|	2g-r-b	
Dicot/Dry	0.35/	0.43/	0.22/	–0.08/	0.21/	14.95/	0.29/	86/42*
Sharpsburg Soil	0.43*	0.23*	0.34*	0.20*	–0.10*	–2.64*	–0.30*	
Dicot/Wet	0.36/	0.43/	0.22/	–0.08/	0.21/	14.96/	0.29/	86/51*
Sharpsburg Soil	0.37	0.26*	0.38*	0.12*	–0.13*	–6.18*	–0.25*	
Dicot/Dry	0.36/	0.43/	0.22/	–0.08/	0.21/	14.96/	0.29/	86/25*
Steinauer Soil	0.39*	0.31*	0.30*	0.08*	0.01*	0.06*	–0.06*	
Dicot/White	0.36/	0.43/	0.22/	–0.08/	0.21/	14.96/	0.29/	86/8*
Corn Residue	0.47*	0.28*	0.26*	0.19*	0.03*	0.12*	–0.16*	
Dicot/White	0.35/	0.43/	0.22/	–0.08/	0.21/	14–95/	0.29/	86/11*
Wheat Residue	0.53*	0.27*	0.20*	0.27*	0.07*	0.26*	–0.20*	
Dicot/Yellow	0.36/	0.43/	0.22/	–0.08/	0.21/	14.96/	0.29/	86/17*
Corn Residue	0.60*	0.27*	0.13*	0.33*	0.14*	0.44*	–0.19*	
Dicot/Yellow	0.36/	0.43/	0.22/	–0.08/	0.21/	14.96/	0.29/	86/13*
Wheat Residue	0.63*	0.24*	0.13*	0.39*	0.11*	0.32*	–0.28*	
Monocot/Dry	0.40/	0.46/	0.14/	–0.06/	0.31/	20.62/	0.38/	72/42*
Sharpsburg Soil	0.43	0.23*	0.34*	0.20*	–0.10*	–2.64*	–0.30*	
Monocot/Wet	0.40/	0.46/	0.14/	–0.06/	0.31/	20.62/	0.38/	72/61*
Sharpsburg Soil	0.37	0.25*	0.38*	0.12*	–0.13*	–6.18*	–0.25*	
Monocot/Dry	0.40/	0.46/	0.14/	–0.06/	0.31/	20.62/	0.38/	72/25*
Steinauer Soil	0.39	0.31*	0.30*	0.08*	0.01*	0.06*	–0.06*	
Monocot/White	0.40/	0.46/	0.14/	–0.06/	0.31/	20.62/	0.38/72/8*	
Corn Residue	0.47*	0.28*	0.26*	0.19*	0.03*	0.12*	–0.16*	
Monocot/White	0.40/	0.46/	0.14/	–0.06/	0.31/	20.62/	0.38/	72/11*
Wheat Residue	0.53*	0.27*	0.20*	0.27*	0.07*	0.26*	–0.20*	
Monocot/Yellow	0.40/	0.46/	0.14/	–0.06/*	0.31/	20.621	0.38/*72/17	
Corn Residue	0.60	0.27	0.13	0.33	0.14	0.44	–0.19	
Monocot/Yellow	0.40/	0.46/	0.14/	–0.06/	0.31/	20.62/	0.38/	72/13*
Wheat Residue	0.63*	0.24*	0.13*	0.39*	0.11*	0.32*	–0.28*	

* Mean values are significantly different from each other at the 0.05 level of significance.

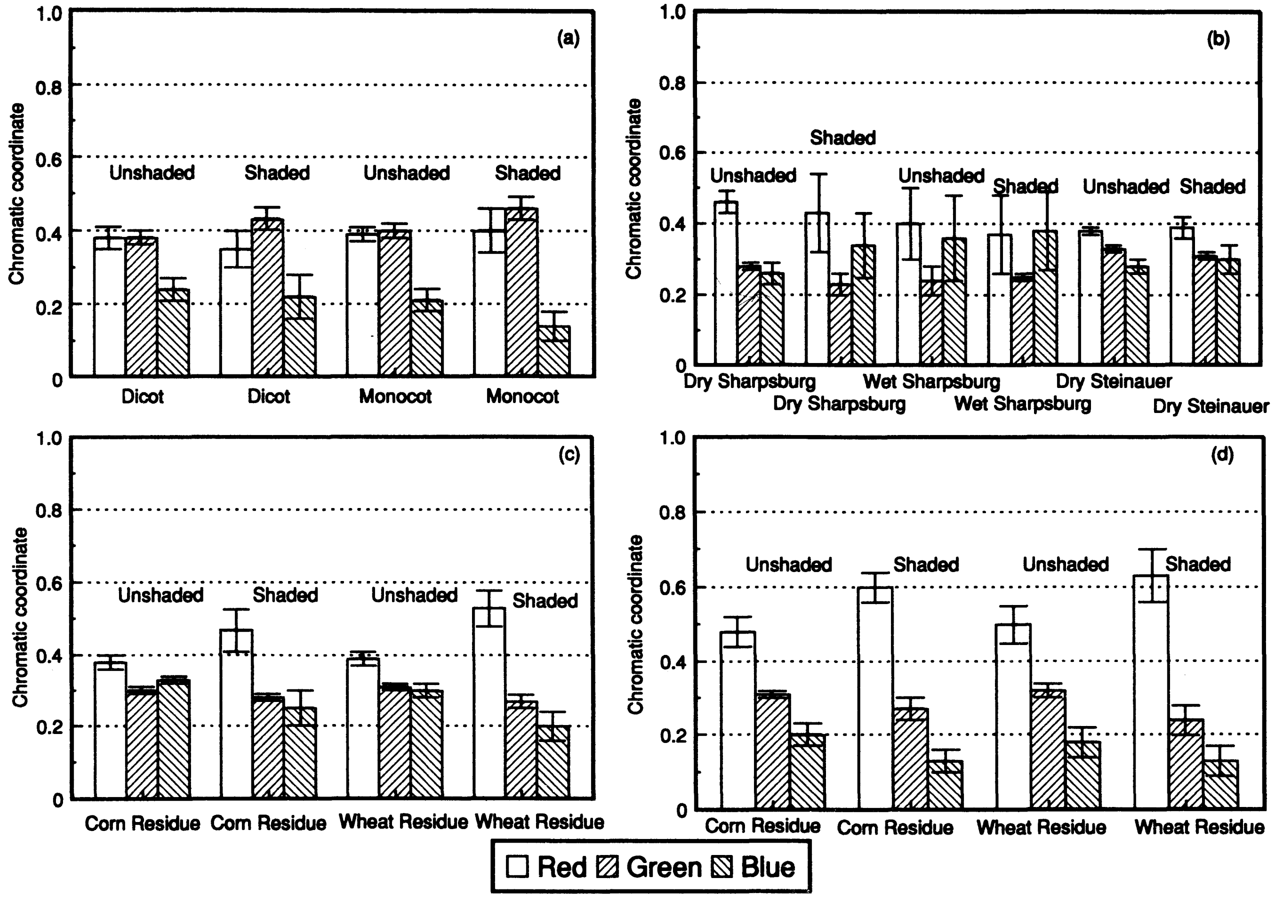

Figure 8–Comparison of chromatic coordinates for improving optical contrast in (a) plants, (b) soils, (c) white residue, and (d) yellow residue. Vertical lines represent 1 standard deviation from the mean.

of nonplant surfaces were always less than those of plant surfaces. Most soil surfaces had higher blue chromaticities than the plant surfaces. Yellow and white residue had a blue chromaticity that was nearly the same as plant surfaces.

Figure 9 shows each chromatic coordinate displayed as a black-and-white image. The green coordinate image shows some promise for distinguishing between plants and their background (fig. 9c, d). The plants appeared brighter than the soil/residue background in these images. However, there is no guarantee that the green coordinate will always classify the plants as they age. The red and blue coordinate images do not work as well. The plants usually have lower values (darker) than the background. This sometimes resulted in a contrasting darker plant image over a lighter background (fig. 9a, b, e, f).

Use of Contrast Indices for Classification

Mean contrast indices values and their variance are given in tables 1 to 4 and figures 10 to 12. Most indices for plant surfaces were significantly different from nonplant surfaces (table 1). There was no difference between dicots and monocots. Index values for shaded and unshaded surfaces were generally significantly different from one another. The r-g index for plant leaves had values that were often negative, while soils and residues had positive values (fig. 10a, b, c, d). The opposite was true with the g-b index (fig. 10). When these indices were combined as a ratio

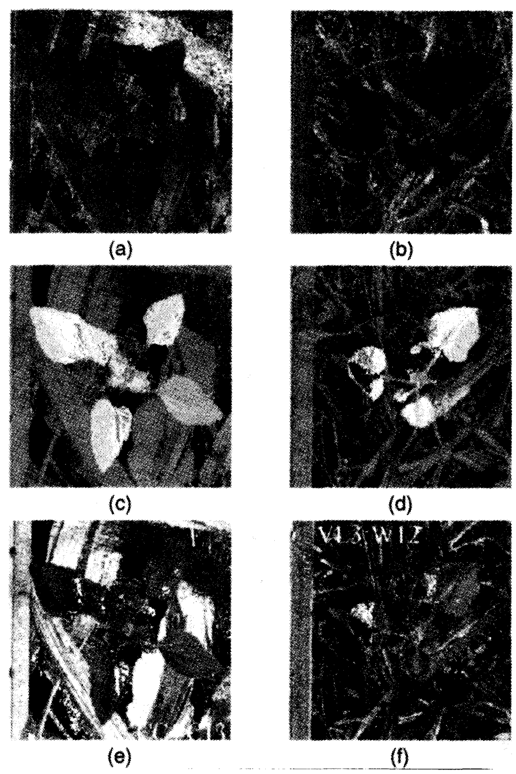

Figure 9–Red (a, b), green (c, d), and blue (e, f) chromatic coefficients of a cocklebur in corn residue (left) and a velvetleaf in wheat residue (right).

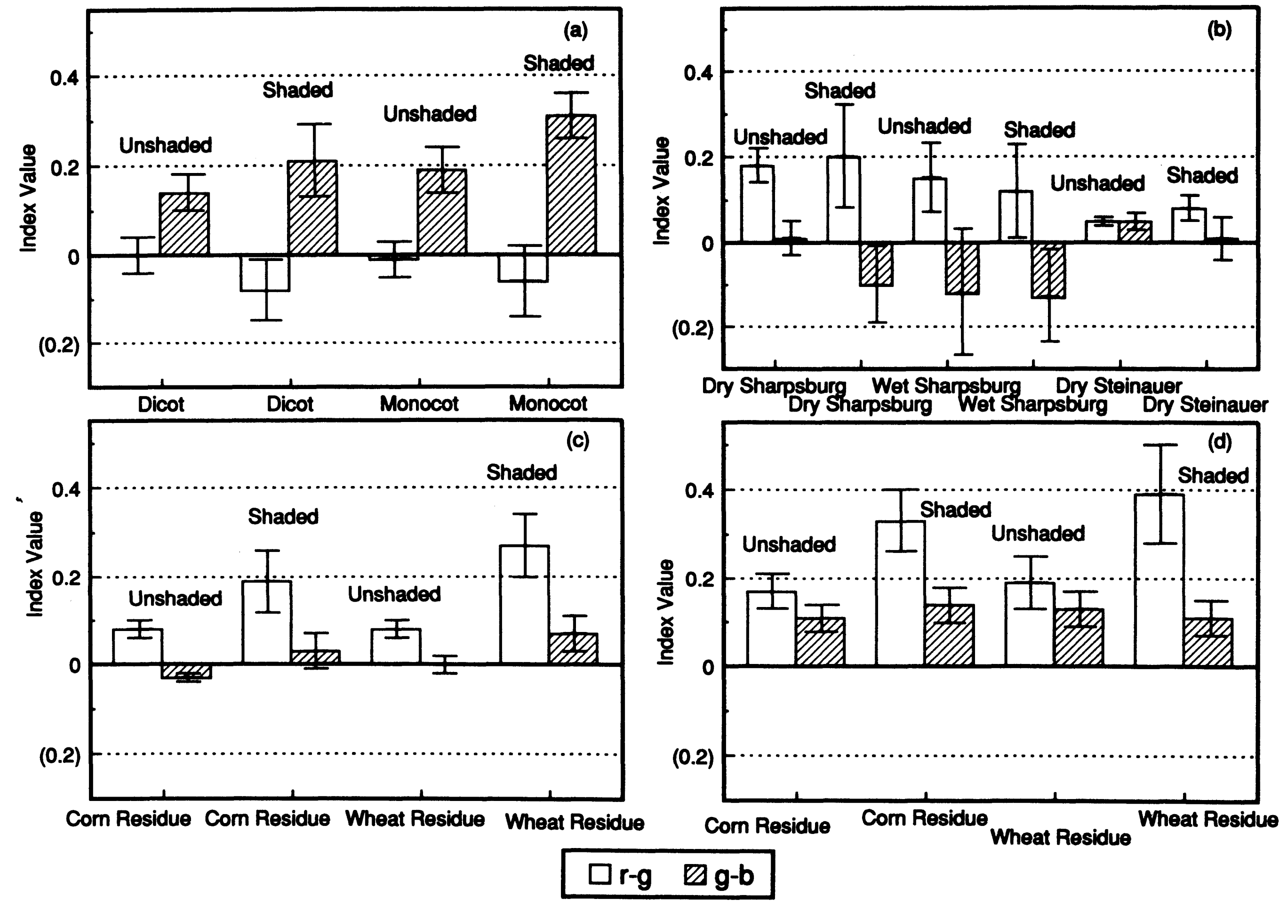

Figure 10–A comparison of the r-g and g-b contrast indices for (a) plants, (b) soils, (c) white residue, and (d) yellow residue. Vertical lines represent 1 standard deviation from the mean.

index (g-b)/|r-g|, better contrast between the plant and background was obtained, but with increased standard deviation (fig. 11). The (g-b)/|r-g| and 2g-r-b contrast indices for the monocots were not consistently higher or lower than dicots for shaded and unshaded surfaces (table 2). The standard deviation of the 2g-r-b index for both plant material and nonplant material was lower than with the (g-b)/|r-g| index (fig. 12). Plant material had significantly higher values of this index than with any other field surface (tables 3 and 4).

Modified Hue

Modified hue distinguished the plant from its background with good success (tables 1 to 4; fig. 13). However, the standard deviation of modified hue for the soils was quite high. This was primarily due to the black or grayish color of

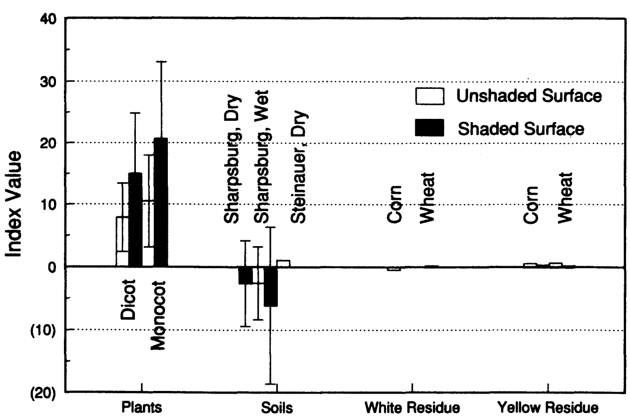

Figure 11–Identifying plants, soils, and residue using the (g-b)/|r-g| index. Vertical lines represent 1 standard deviation from the mean.

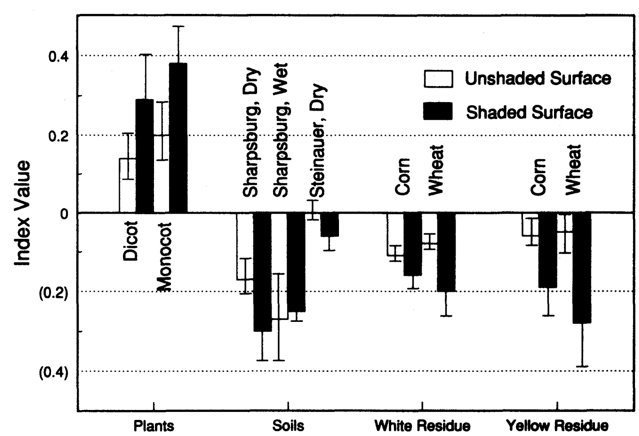

Figure 12–Identifying plant, soil, and residue surfaces using the 2g-r-b index. Vertical lines represent 1 standard deviation from the mean.

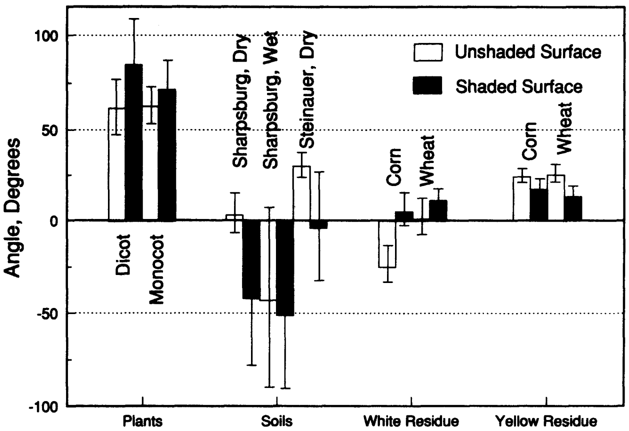

Figure 13–Identifying plant, soils, and residue using modified hue from the HSI color model. Vertical lines represent 1 standard deviation. from the mean.

the soil. The color of the soils is near the brightness axis, therefore, hue is quite variable (fig. 4). The modified hues of shaded and unshaded surfaces were significantly different for most surfaces (table 1). The modified hue could not distinguish between monocots and dicots.

The best contrast results between plant and background were obtained using either the 2g-r-b index or the green chromatic coordinate. The modified hue also significantly distinguished plants from their background. The 2g-r-b index and the green chromatic coordinate were easier to compute than the modified hue, resulting in faster image processing. This would be an important consideration for field applications.

Contrast Indices Applied to Entire Images

All of the coordinate indices improved the contrast of the test areas of interest between the plant and background, at the 0.05 significance level. Figures 14 and 15 represent the results when the indices were applied to the entire image. The r-g index produced a dark plant image with a slightly lighter gray residue (figs. 14a, 15a). The difference in gray levels between the plant and the background was not significant. The black "holes" within the leaves of these images were the result of using 8-bit images (note that zero is obtained for all indices when a gray level is introduced). When the g-b index was used, the plants were bright, residue was darker, and soil was black (fig. 14b, 15b). This index, though it did not eliminate the background from the plant image, provided adequate binary contrast.

The background was completely removed from the image when the indices (g-b)/|r-g| and 2g-r-b were used (figs. 14c, d; 15c, d). However, the leaf on the right side of the plant in figure 14c was also partially removed. That leaf was yellow-green, which the index (g-b)/|r-g| failed to identify. However, the 2g-r-b index did recognize this leaf as part of the plant and it appears as a very dark gray in figure 14d. The 2g-r-b index provided the best contrast between plant and nonplant surfaces.

Hue and Modified Hue Applied to Entire Images

The hue for plants was very different from the hue of other surfaces. However, intensity was between that of the soil and residue (fig. 14f). The modified hue had a light gray color for plants, while the background was very dark, although the texture associated with the residue was faintly visible (fig. 14e). This allowed the use of a single threshold gray-level value for distinguishing green plant material from soil/residue background. The modified hue had supporting statistical values, although not as high as with the 2g-r-b index or the green chromatic coordinate. Therefore, the best methods distinguishing plant surfaces from non-plant surfaces in color images were either the 2g-r-b index or the green chromatic coordinate.

Conclusions

Photographic slide images of plants against a natural background of soils and residues were analyzed for color as a method of distinguishing the two. The non-normalized RGB coordinates were found very inadequate for distinguishing plant surfaces from nonplant surfaces. Non-normalized coordinates had large variances depending upon camera and scanner settings, as well as the direction and amount of incident light.

The red, green, and blue normalized chromatic coordinates were useful for consistently quantifying color of each surface. Most shaded and unshaded surfaces of the same type had significantly different chromatic coordinates. The green chromatic coordinate was

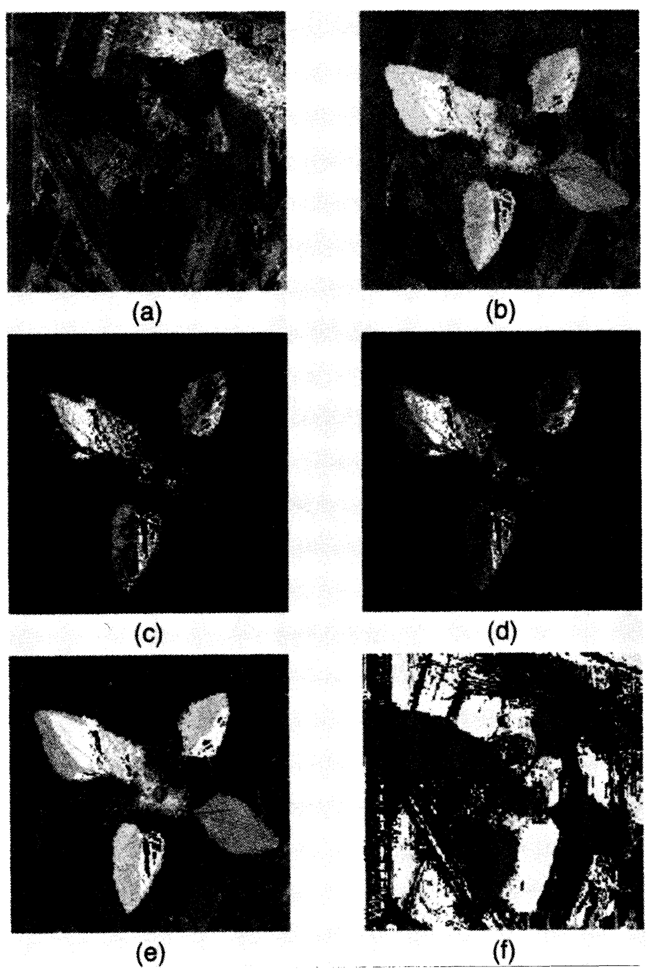

Figure 14–Contrast indices for a cocklebur in corn residue (a) r-g, (b) g-b, (c) (g-b)/(r-g), (d) 2g-r-b, (e) modified hue, and (f) hue.

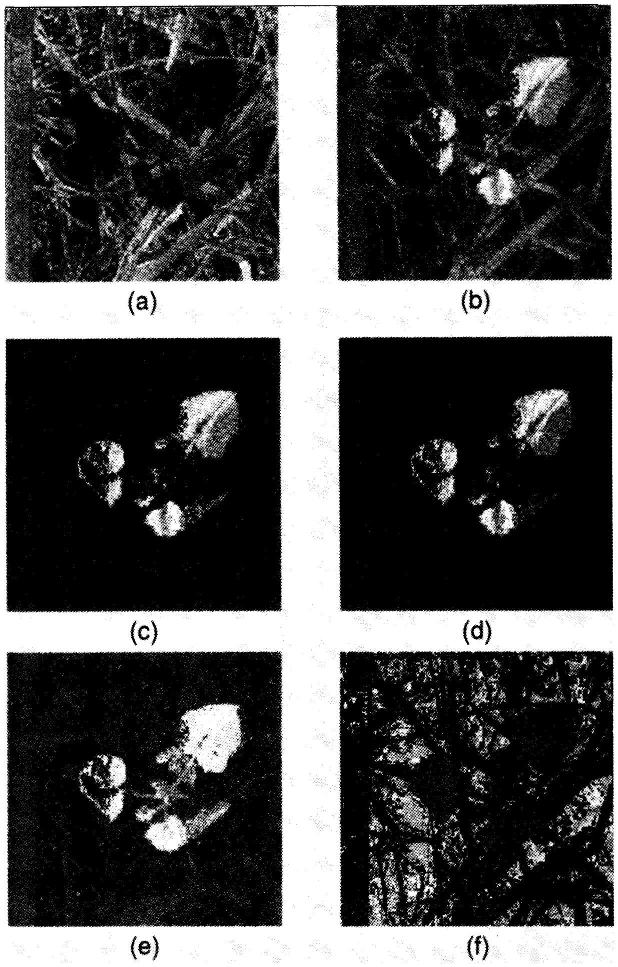

Figure 15–Contrast indices for a velvetleaf in wheat residue. (a) r-g, (b) g-b, (c) (g-b)/(r-g), (d) 2g-r-b, (e) modified hue, and (f) raw hue.

significantly higher for a plant than for other surfaces. The green chromatic coordinate of monocots was significantly different from the dicots, but not in a consistent manner. Leaf "hole" pixels (created by the conversion from 24-bit to 8-bit images) created only minor inconsistencies. However, 24-bit processing technology is improving steadily every day and should be used in future studies.

The best separation of plants and background occurred with the modified hue or the 2g-r-b contrast index. Modified hue was the most computationally intense, but this is only a short-term disadvantage with faster processors coming on the market. Color was unable to distinguish monocots from dicots. However, the resultant near-binary image is the next step to canopy shape feature analysis.

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